

Question	Answer	Mark	Mark scheme	Additional guidance
14	No (supported)	P1 P1 P1 P1 C1	for $P(OO) = \frac{5}{9} \times \frac{4}{8} (= \frac{20}{72})$ or $P(OE) = \frac{5}{9} \times \frac{4}{8} (= \frac{20}{72})$ or $P(EO) = \frac{4}{9} \times \frac{5}{8} (= \frac{20}{72})$ or $P(EE) = \frac{4}{9} \times \frac{3}{8} (= \frac{12}{72})$ for $P(OO) = \frac{5}{9} \times \frac{4}{8} (= \frac{20}{72})$ and $P(EE) = \frac{4}{9} \times \frac{3}{8} (= \frac{12}{72})$ OR for $P(OE) = \frac{5}{9} \times \frac{4}{8} (= \frac{20}{72})$ and $P(EO) = \frac{4}{9} \times \frac{5}{8} (= \frac{20}{72})$ and $P(EE) = \frac{4}{9} \times \frac{3}{8} (= \frac{12}{72})$ for a process to find probability of sum being even, eg $P(OO) + P(EE) = \frac{5}{9} \times \frac{4}{8} + \frac{4}{9} \times \frac{3}{8} (= \frac{32}{72})$ for a process to work with probability of product being even, eg $P(EO) + P(OE) + P(EE) = \frac{4}{9} \times \frac{5}{8} + \frac{5}{9} \times \frac{4}{8} + \frac{4}{9} \times \frac{3}{8} (= \frac{52}{72})$ or $1 - P(OO) = 1 - \frac{5}{9} \times \frac{4}{8} (= \frac{52}{72})$ for No supported by correct probabilities, eg $\frac{32}{72}$ and $\frac{52}{72}$ SC B2 for $\frac{41}{81}$ and $\frac{56}{81}$ and No SC B1 for $\frac{41}{81}$ and $\frac{56}{81}$ with no decision or incorrect decision	Accept equivalent probabilities throughout Sample space diagram or listing: Award P3 for $P(\text{sum even}) = \frac{32}{72}$ or $P(\text{product even}) = \frac{52}{72}$, P4 for both